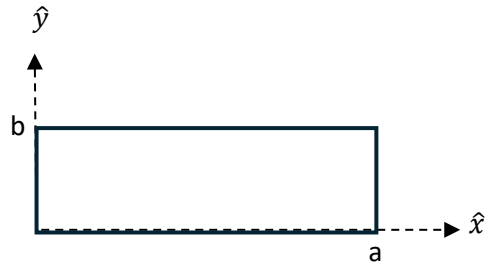
Physics 2 Problem Solving – Chapter 48 Magnetic Flux and EMI

1) A rectangular wire loop is shown below with length "a" and width "b". The magnetic field inside the loop is pointing out of the page (+ $\hat{z}$) direction and is change along the x-axis a given, where c is a constant and t is time.

$$\vec{B} = \left(\frac{cx^2}{t}\right)\hat{z}$$



A) Draw the relevant variables on the diagram to solve the problem.

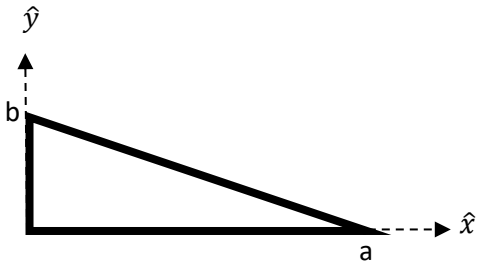
B) Determine the equation for the magnetic flux through the inside of the rectangle.

$$Answer:\ \Phi_m = \frac{cba^3}{3}$$

C) Determine the equation for the induced potential (V_{enf}) in the rectangular wire.

$$Answer:\ V_{emf} = \frac{Ncba^3}{3t^2}$$

2) A triangular wire loop is shown below with length "a" and heigth "b". The magnetic field inside the loop is pointing out of the page (+ $\hat{z}$) direction. Must use calculus for credit



A) Draw the relevant variables on the diagram to solve the problem.

B) Determine the equation for the magnetic flux through the inside of the triangle.

$$Answer:\ \Phi_m = \left|\vec{B}\right|\left(\frac{ab}{2}\right)$$

C) Determine the equation for the induced potential (V_{enf}) in the rectangular wire if the magnetic field is changing with time given by the equation below.

$$Answer:\ \vec{B} = \left(\frac{abNhB_o}{4t^{1/2}}\right)\hat{z}$$